Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State that you are treating air resistance as small enough to ignore.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your answer. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Warm-up: bring back an earlier idea

Outside a spherical source, field at r is **75 N/kg**. **Find field** at **5** r and the force on **0.7 kg** there.

Word help: field = a description of an influence at each location

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: a_x = **0**; a_y = -g; v_x = constant.
Upward positive; g = **10 m/s²**. Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at **3 m/s** and falls for **2 s**. Horizontal travel is **3×2=6 m**. With g=**10 m/s²** and no initial vertical velocity, its final vertical velocity is **20 m/s** down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after **2 s**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

One ball is dropped and another launched horizontally from the same height at the same instant. **Compare fall times** and explain.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball rolls horizontally off a platform at **8 m/s**. It lands **1.5 s** later. **Label the diagram** and **find horizontal displacement**, vertical displacement and both final velocity components. Take upward as positive, use g = **10 m/s²** and ignore air resistance.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

Independent projectile directions

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a larger horizontal speed holds the ball up longer. **Explain which force or acceleration** would be needed for that statement.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

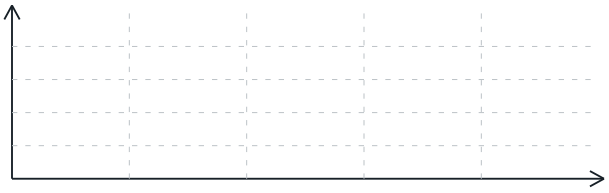
06 Review: connect an earlier idea

For x_i=**4±1 m**, v=**6±0.5 m/s** and exact t=**8 s**, find the central final position and conservative interval. **Name a specific check of** your method.
Label the measured positions and the range each allows.

Word help: conservative bound = a limit chosen to include every allowed case

Sentence starter: I found … [unit]. This value means … in this situation.

Position and uncertainty (m)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A projectile's velocity is (**6**,**-8**) m/s. **Find speed**. What information is still needed to **find its position**? **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: I found … [unit]. This value means … in this situation.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A projectile has initial velocity (**6**, **8**) m/s; use upward positive and g=**10 m/s²**. **Find velocity after 1 s** and **explain whether** its horizontal motion depends on its vertical velocity. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3